RAMAIAH
Institute of Technology

Programme: B E – Department of Artificial Intelligence and Data Science

Internal Assessment – I

TERM : 15/04/2024 to 27/07/2024	COURSE NAME: Data Communication and Networking
DATE : 30-05-2024 TIME: 9:30AM- 10:30AM	COURSE CODE : AD42
MAX MARKS: 30	PORTIONS : L1-L15

Mobile Phones are banned

Instructions to Candidates: **Answer Any Two Full Questions.** Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Explain the following with neat diagrams: 3. Point to Point and Multipoint connection 4. Simplex and full duplex mode	L2	CO1	04
b	A signal with data stream 01010101 is transmitted. Draw the line coding of the following: i) NRZ-I ii) Manchester iii) AMI	L3	CO2	06
c	Explain Nyquist bit rate and Shannon Capacity with suitable formula.	L2	CO2	05
2.a	Differentiate between Mesh and Star topology using suitable example and diagrams.	L3	CO1	05
b	Describe the responsibilities of Physical and Data link layer in OSI reference model	L2	CO1	05
c	What is the theoretical capacity of a channel in each of the following cases? iii) Bandwidth= 20 KHz, $SNR_{dB} = 40$ iv) Bandwidth= 1MHz, $SNR_{dB} = 20$.	L3	CO2	05
3.a	With a neat diagram, explain TCP/IP protocol suite.	L2	CO1	08
b	Illustrate the three causes of transmission impairment in signal transmission.	L2	CO2	07

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create

RAMAIAH
Institute of Technology

Programme: B E – Department of Artificial Intelligence and Machine Learning

Internal Assessment – I

TERM : 15/04/2024 to 27/07/2024	COURSE NAME: Data Communication and Networking
DATE : 30-05-2024TIME:09:30AM-10:30AM	COURSE CODE : AI42
MAX MARKS: 30	PORTIONS : L1-L15

Mobile Phones are banned

Instructions to Candidates: **Answer any two questions.** Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Discuss the steps involved in transferring a web page from client to serverusing HTTP with non-persistent connection.	L2	CO1	5
b	Consider distributing a file of F = 15 Gbits to N peers. The server has an upload rate of u_s = 30 Mbps, and each peer has a download rate of d_i = 2 Mbps and an upload rate of u=700 kbps and 2 Mbps. For N = 1000, prepare a chart giving the minimum distribution time for each of the combinations of N and u for both client-server distribution and P2P distribution.	L3	CO1	5
c	Discuss TCP round trip time and timeout interval with suitable formulas.	L2	CO2	5
2.a	Draw TCP segment structure and explain the flag fields.	L2	CO2	6
b	Analyse with suitable example TCP slow start and congestion avoidance mechanism.	L4	CO2	5
c	Explain with an example various DNS resource records.	L2	CO1	4
3.a	Consider e-commerce site that wants to keep a purchase record for each of its customers. Analyse with neat diagram how this can be done with cookies.	L4	CO1	4
b	Illustrate connection oriented multiplexing and de-multiplexing with neat figure.	L3	CO2	5
c	Distinguish between mesh topology and star topology with its advantages and disadvantages.	L4	CO1	6

***L1** – Remember, **L2** – Understand, **L3**- Apply, **L4**- Analyze, **L5**-Evaluate, **L6**-Create

Course Outcomes meant to be assessed by the first IA Test:

1. Describe the fundamentals of data communications and various application layer protocols used by TCP/IP reference model. (PO-1, 2, 3, 4, 10, PSO1).
2. Differentiate between connection oriented and connection less services of transport layer (PO-1, 2, 3, 4, 10, PSO1

Programme: B E – Department of Artificial Intelligence and Data Science

Internal Assessment – I

TERM : 15/04/2024 to 27/07/2024	COURSE NAME: Data Communication and Networking
DATE : 15-07-2024 TIME:12:00PM- 1:00PM	COURSE CODE : AD42
MAX MARKS: 30	PORTIONS : L1-L15

Mobile Phones are banned

Instructions to Candidates: **Answer Any Two Full Questions.**

Marks: **14x2=28**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Illustrate the process of Quantization in Pulse Code Modulation by considering eight quantization levels, sample normalized amplitude values: 7.5, 19.7, 11.0, -5.5, -11.3, -6.0 and sample amplitudes between -20V and +20 V. | Transmission Impairment	L2	CO2	08
b	Illustrate with examples the four levels of addresses in TCP/IP reference model	L3	CO1	06
2.a	With a neat diagram, explain TCP/IP reference model in detail.	L2	CO1	08
b	Calculate maximum capacity of a channel using suitable theorem for following cases: a) A telephone line with SNR_{dB} ratio of 40dB and an audio bandwidth of 3kHz. b) A satellite TV channel with SNR ratio of 100 and a video bandwidth of 1MHz	L3	CO2	06
3.a	Draw the mesh and bus topology for 5 nodes. Compare the two topologies in terms of: 1. Type and Number of cables required. 2. Number of connectors required. 3. If one node is added and removed, then what changes take place in topology.	L3	CO1	06
b	Consider that the signal with data stream 10011010 is transmitted over transmission medium. Draw the graphs for the following line coding techniques: i) Unipolar NRZ ii) Polar NRZ-I iii) Differential Manchester iv) Pseudo ternary.	L2	CO2	08

* **L1** – Remember, **L2** – Understand, **L3**- Apply, **L4**- Analyze, **L5**-Evaluate, **L6**-Create

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Institute of Technology

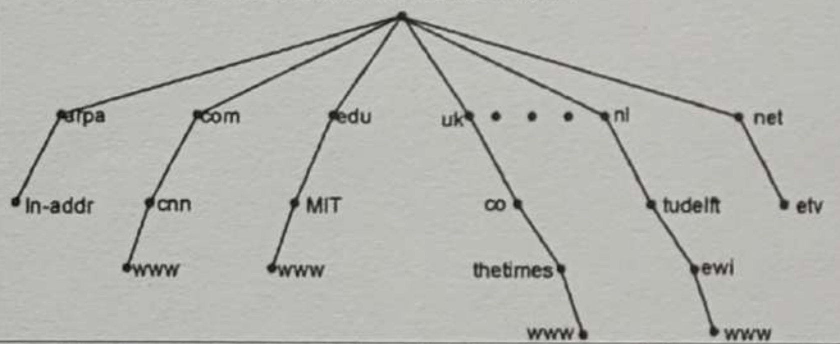
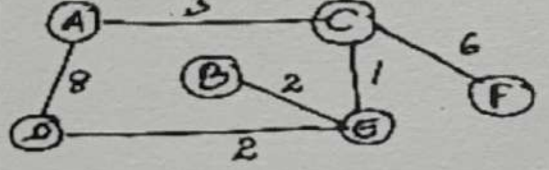
Programme: B E – Department of Artificial Intelligence and Data Science

TERM : 15/04/2024 to 27/07/2024	COURSE NAME: Data Communication and Networking
DATE : 23-07-2024 TIME: 9:30AM- 10:30AM	COURSE CODE : AD42
MAX MARKS: **30**	PORTIONS : L16-L38

Mobile Phones are banned

Internal Assessment – II

Instructions to Candidates: **Answer Any Two Full Questions**. Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	An organization is granted a block of addresses with the beginning address as 123.22.21.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 15 addresses, one subblock of 50 addresses, and one subblock of 140 addresses. Design the subblocks.	**L3**	**CO3**	**05**
b	Explain with neat diagrams the first two phases involved in connection oriented TCP protocol.	**L2**	**CO4**	06
c	Consider the following domain name space tree. Identify the following domain names belong to FQDN/PQDN/Inverse along with generic/country domain.Justity 1. www.cnn 2. www.thetimes.co.uk. 3. etv.net 4. Write the inverse domain for 123.23.145.20 	**L3**	**CO5**	04
2.a	Consider the network given below, Find the shortest path from node A to all other nodes using Dijkstra's algorithm. Draw the shortest path trees and shortest path table for all the iterations: 	**L3**	**CO4**	**05**
b	Explain with neat diagram all the fields in IPv4 datagram packet header.	**L2**	**CO3**	**06**
c	Explain with suitable diagram Iterative DNS resolvers.	**L2**	**CO5**	04
3.a	Illustrate the drawbacks of Distance Vector Routing.	**L3**	**CO4**	**05**
b	Consider the public IP address as 200.24.5.8 and private IP address as 172.18.3.1, explain with diagram the steps involved suing Network Address Translation (NAT) for connecting private network to Internet.	**L3**	**CO3**	05
c	Explain suitable diagram, Slow start, Exponential Increase phase of congestion control in TCP	**L2**	**CO5**	05

CIE: Internal Assessment Details

Revision as per format MSRIT .F702 Rev. No. 2

Internal Assessment Question Paper – 1
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)

Department of AI&ML/AI&DS **Programme : B.E** **Term: May-Aug 2023**
Course: Data Communication & Networking **Course Code: AI42/AD42** **CIE: Test 1 Sem: IV**
Max Marks: 30 **Date: 13/06/2023**
Portions for Test: (L01 to L18) **Time: 2.00- 3.00**

Instructions to Candidates:

- **Question 1 is compulsory. Answer any 2 full questions.**
- **Each question carries 15 marks. Mobile are strictly banned**

Sl #	Question	Marks	Bloom's Level	CO Mapping
1	a. Draw the general format of an HTTP request message and give an example for request and response messages using conditional Get.	6	L2	CO1
	b. Identify various fields in the TCP segment structure and explain any four flag fields.	5	L3	CO2
	c. Discuss various DNS resource records with an example.	4	L2	CO1
2	a. Show the layers in the TCP/IP Protocol Suite and explain the services provided by the network layer.	5	L2	CO1
	b. With a neat diagram identify the sequence of TCP states visited by a client TCP	5	L3	CO2
	c. With suitable formulas explain file distribution using P2P and client-server approach.	5	L4	CO1
3	a. With suitable formulas explain what happens if the network layer delivers data faster than the application layer removes data from socket buffers.	5	L4	CO1
	b. Distinguish between mesh topology and star topology.	5	L4	CO1
	c. Illustrate with an example Connectionless demultiplexing at the transport layer.	5	L2	CO2

Course Outcomes meant to be assessed by the first IA Test:

1. Describe the fundamentals of data communications and various application layer protocols used by TCP/IP reference model. (PO-1, 2, 3, 4, 10, PSO1).
2. Differentiate between connection oriented and connection less services of transport layer (PO-1, 2, 3, 4, 10, PSO1).

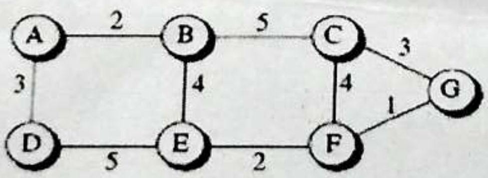
CIE: Internal Assessment Details

Internal Assessment Question Paper – 2
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)

Department of AI&ML/AI&DS **Programme: B.E** **Term: May-Aug 2023**
Course: Data Communication & Networking Course Code: AI42/AD42 CIE: Test 2 Sem: IV
Max Marks: 30 **Date: 31/07/2023**
Portions for Test: (L19 to L24) **Time: 2.00- 3.00**

Instructions to Candidates:

- **Question 1 is compulsory. Answer any 2 full questions.**
- **Each question carries 15 marks. Mobile are strictly banned**

Sl#	Question	Marks	Bloom's Level	CO Mapping
1	a. With neat diagram explain the process of encoding and decoding in CRC.	05	L2	CO4
	b. An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have three subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.	05	L3	CO3
	c. Illustrate three causes of signal impairment with an example.	05	L2	CO5
2	a. Draw the flow diagram to show the working of CSMA/CD.	04	L2	CO4
	b. Differentiate between Nyquist Bit Rate and Shannon's Capacity to calculate the data rate limits.	05	L4	CO5
	c. solve the following i. We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What is the appropriate bit rate and signal level? ii. The signal-to-noise ratio is often given in decibels. Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. calculate the theoretical channel capacity.	06	L3	CO5
3	a. Identify the need for fragmentation at network layer. Consider a datagram of 3000Bytes (20 bytes of header +2980 bytes of IP payload) reached at the router and must be forwarded to the link with 500 bytes. How many fragments will be generated and identify the MF bit and offset value for each fragment.	05	L3	CO3
	b. Consider the following network with indicated link cost, use Dijkstra's shortest path algorithm to compute the shortest path from node A to all nodes in the network and draw the least cost tree.  	05	L3	CO3
	c. Differentiate between character oriented and bit-oriented framing with suitable example.	05	L2	CO4

Course Outcomes meant to be assessed by the second IA Test:

1. Solve problems of IP addressing and routing using various routing protocols and algorithms. (PO-1, 2, 3, 4, 10, PSO1).
2. Illustrate error control and media access control protocols of data link layer (PO-1, 2, 3, 4, 10, PSO1).
3. Discuss different types of data transmission techniques. (PO-1, 2, 3, 4, 10, PSO 1).

AI42/AD42

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BACKLOG SUBJECT EXAMINATIONS – FEBRUARY 2024

Program	:	B.E. – Artificial Intelligence and Data Science / Artificial Intelligence and Machine Learning	Semester	:	IV
Course Name	:	Data Communication and Networking	Max. Marks	:	100
Course Code	:	AI42 / AD42	Duration	:	3 Hrs

Instructions to the Candidates:
- Answer one full question from each unit.

UNIT – I

1.	a)	With the help of an example, describe a nonpersistent connection.	CO1	(10)
	b)	Define FTP. List out the steps used to map the hostname to an IP address.	CO1	(10)
2.	a)	Describe the format used for the request and response messages in HTTP.	CO1	(10)
	b)	Illustrate DNS message format using a block diagram.	CO1	(10)

UNIT – II

3.	a)	With a diagram, describe the process of data transfer in TCP. Draw the frame format for TCP segment.	CO2	(10)
	b)	Discuss round trip time estimation and time out interval with suitable formulas.	CO2	(10)
4.	a)	Enumerate user datagram protocol packets. Calculate the time required by TCP before the sequence number goes back to zero, if sending data speed is 1 Mbyte/s and the sequence number starts at 7000.	CO2	(10)
	b)	Illustrate congestion avoidance mechanism with neat figure.	CO2	(10)

UNIT – III

5.	a)	What are the restrictions imposed on classless address blocks? An Organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets. (i) Find the subnet mask (ii) Find the number of addresses in each subnet (iii) Find the first and last address of subnet 1 (iv) Find the first and last address of subnet 32 (v) Find the network address of subnet 3.	CO3	(10)
	b)	Illustrate fragmentation and reassembling in IP with example.	CO3	(10)
6.	a)	Describe the link state routing protocol with example.	CO3	(10)
	b)	Define IPV4 Addresses the process for creating a link-state database.	CO3	(10)

UNIT – IV

7.	a)	Given the Data word 1001 and the divisor 1011, generate the codeword at the sender site using CRC polynomial division. Assuming no error calculate the syndrome at the receiver site.	CO4	(10)
	b)	With a flow diagram, illustrate the Stop and wait protocol.	CO4	(10)
8.	a)	Describe CSMA/CA using a flow diagram.	CO4	(10)
	b)	With the help of a diagram, discuss CSMA/CD.	CO4	(10)

UNIT-V

9.	a)	Describe the digital to analog conversation with example.	CO5	(10)
	b)	Explain delta modulation technique with neat figure.	CO5	(10)
10.	a)	Illustrate Time-Division Multiplexing. An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?	CO5	(10)
	b)	Illustrate Components of PCM encoder using block a diagram.	CO5	(10)

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RAMAIAH Institute of Technology

Dept. of Computer Science and Engineering (AI & ML) & Computer Science and Engineering (Cyber Security)

Programme: B E – Computer Science and Engineering (AI&ML) and CSE (Cyber Security)

Internal Assessment – I

TERM: 15th April 2024 – 27th July 2024	COURSE NAME: Data Communication and Networking
DATE: **30-05-2024** TIME: **9.30-10.30**	COURSE CODE: CY42/CI42
MAX MARKS: **30**	PORTIONS: L1 to L19

Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.** Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Consider a multinational corporation with offices in five different countries. Each office needs to be interconnected for seamless communication, file sharing, and collaborative work in a mesh topology. How many connections are needed for mesh network? Compare it with a star topology, where all offices are connected to a central headquarters or data centre.	L3	CO1	5
1.b	Explain architecture and functionality associated with each layer of OSI model and compare it with TCP/IP protocol suite.	L4	CO1	10
2.a	Draw the line coding graph for Unipolar, RZ, NRZ-L, NRZ-I, Manchester, Differential Manchester, AMI and Pseudo ternary schemes for the following data stream: 10011010.	L2	CO2	8
2.b	Consider a network engineer responsible for managing data transmission between two offices of a multinational corporation located in different cities. Node A has a data word "101001111" representing a segment of the financial report. Before transmitting this data to Node B, it needs to be encoded with CRC using the divisor "10111". Calculate CRC.	L3	CO3	4
2.c	Consider a noisy channel with $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. Calculate the capacity of the channel.	L3	CO2	3
3.a	Consider Alice wants to securely transmit a 4-bit message 1101 to Bob over an unreliable communication channel. She wants to ensure that the message reaches Bob without any errors. To achieve this, she decides to encode the message using Hamming code before transmission. Calculate the hamming code to ensure error detection during transmission.	L3	CO3	6
3.b	Describe transmission impairment and explain the causes for it.	L2	CO2	6
3.c	Consider a binary code that contains 4 valid codewords as follows: 00000,01011,10101, 11110.Calculate d_{min} and calculate the maximum number of erroneous bits that can be corrected by the d_{min}.	L3	CO3	3

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SEMESTER END EXAMINATIONS – AUGUST 2024

Program	:	**B.E. – CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**IV**
Course Name	:	**Data Communication and Networking**	**Max. Marks**	:	**100**
Course Code	:	**CY42 / CI42**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Enumerate the functions associated with each layer of TCP/IP reference model.	CO1	(08)
	b)	With necessary diagrams, explain different types of network topology.	CO1	(08)
	c)	Differentiate between host to host delivery and end to end delivery of data.	CO1	(04)
2.	a)	Explain the fundamental components of data communication systems with an example for each.	CO1	(06)
	b)	The number of duplex mode liners in a mesh topology is n(n-1)/2. Justify the statement. List the merits and demerits of mesh topology.	CO1	(08)
	c)	Differentiate between the following: i) Encapsulation and Decapsulation ii) logical and physical addressing.	CO1	(06)

UNIT - II

3.	a)	Draw the graph of unipolar NRZ, NRZ-I, Polar RZ, Manchester and Differential Manchester for the data stream 1001101010 assuming that the last signal level has been positive.	CO2	(08)
	b)	Define bit rate and baud rate. Explain the line coding schemes with suitable examples.	CO2	(06)
	c)	Illustrate the reasons for transmission impairments in data communication and explain the categories of transmission impairments.	CO2	(06)
4.	a)	What is SNR? How can we calculate SNR_{db} from SNR? For a channel of 1Mhz bandwidth and SNR 63, Calculate the appropriate bit rate and signal level.	CO2	(08)
	b)	What is the total delay for a frame of size 5 million bits that are being sent on a link with 10 routers each router having a queuing time of 2 μs and a processing time of 1 μs. The length of the link is 2000km. The speed of light inside the link is 2x10 8 m/s. The link has a bandwidth of 5 Mbps. Which component of the total delay is dominant? Which one is negligible?	CO2	(06)
	c)	Discuss the use of Nyquist bit rate and Shannon capacity in computing the data rates. The signal-to-noise ratio is often given in decibels. Assume that SNRdb = 36 and the channel bandwidth is 2MHz. Calculate the channel capacity.	CO2	(06)

UNIT - III

5.	a)	Define Cyclic Code and explain with neat block diagrams CRC encoder and decoder.	CO3	(05)
	b)	Consider a network engineer responsible for managing data transmission between two offices of a multinational corporation located in different cities. Node A has a data word "101001111" representing a segment of the financial report. Before transmitting this data to Node B, it needs to be encoded with CRC using the divisor "10111". Calculate Cyclic Redundancy Check.	CO3	(10)
	c)	Compare and contrast flow control and error control with suitable examples.	CO3	(05)
6.	a)	Compare and contrast the Go-Back-NARQ Protocol with Selective-Repeat ARQ with appropriate timing diagrams.	CO3	(10)
	b)	Consider Alice wants to securely transmit a 4-bit message 1101 to Bob over an unreliable communication channel. She wants to ensure that the message reaches Bob without any errors. To achieve this, she decides to encode the message using Hamming code before transmission. Calculate the hamming code to ensure error detection during transmission.	CO3	(06)
	c)	Consider a binary code that contains 4 valid code words as follows: 00000,01011,10101,11110.Calculate d_{min} and calculate the maximum number of erroneous bits that can be corrected by the d_{min}.	CO3	(04)

UNIT- IV

7.	a)	Discuss the relevance of CSMA/CA and explain the strategies used for avoiding collision using a flow diagram.	CO4	(08)
	b)	Discuss different channelization techniques in detail. Calculate chip code for a network with 2 stations and 4 stations using Walsh table W_1 =[1].	CO4	(08)
	c)	Compare CSMA with ALOHA protocol.	CO4	(04)
8.	a)	Discuss the different persistence method and explain CSMA/CD with a neat flow diagram.	CO4	(08)
	b)	In a CSMA/CD network with a data rate of 10 Mbps, the minimum frame size is found to be 512 bits for the correct operation of the collision detection process. What should be the minimum frame size if we increase the data rate to i) 100 Mbps. ii) 1Gbps. iii) 10 Gbps.	CO4	(06)
	c)	Discuss two controlled access protocols for media access.	CO4	(06)

UNIT - V

9.	a)	i) Discuss and differentiate the common physical layer implementation of Standard Ethernet with suitable diagrams. ii) Consider the length of 10Base5 cable is 2500m. If the speed of a propagation in a thick coaxial cable is 200,000,000 m/s, how long does it take for a bit to travel from the beginning to the end of the network? Assume there is 10μs delay in the equipment.	CO5	(08)

	b)	i) The address 43:7B:6C:ED:10:00 has been shown as source address in an Ethernet frame. The receiver has discarded the frame. Give reason. ii) If an Ethernet destination address is 05:01:02:03:04:05, What is the type of the address? Give reason. iii) Show how the address 47:20:1B:2E:08:EE is sent out on line. iv) If an Ethernet destination address is FF:FF:FF:FF:FF:FF, what is the type of the address? Give reason.	CO5	(06)
	c)	Explain the addressing mechanism of IEEE 802.11 protocol.	CO5	(06)
10.	a)	Explain Bluetooth and its architecture with a neat diagram.	CO5	(08)
	b)	Explain Ethernet Frame format with a neat diagram. Consider an Ethernet MAC sublayer receives 32 bytes of data from the upper layer. How many padding must be added to the data?	CO5	(06)
	c)	i) Describe connecting devices based on the layers they operate in a network with a neat diagram. ii) Which one has more overhead, a repeater or a bridge? Defend your answer.	CO5	(06)

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SEMESTER END EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	:	**B.E. - Artificial Intelligence and Machine Learning / Artificial Intelligence and Data Science**	**Semester**	:	**IV**
Course Name	:	**Data Communication and Networking**	**Max. Marks**	:	**100**
Course Code	:	**AI42 / AD42**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Explain TCP/IP protocol suite with a neat figure.	CO1	(06)
	b)	Differentiate between persistent and non-persistent TCP connections used in HTTP.	CO1	(06)
	c)	Discuss DNS records and messages with neat format.	CO1	(08)
2.	a)	Differentiate between GET and Conditional GET.	CO1	(06)
	b)	Explain use of cookies to maintain state information.	CO1	(06)
	c)	Differentiate between mesh and star topology with suitable figure.	CO1	(08)

UNIT - II

3.	a)	Explain with a neat diagram how the transport layer handle the multiplexing and demultiplexing of the data.	CO2	(08)
	b)	Discuss with your own example how UDP checksum provides the error detection.	CO2	(06)
	c)	Explain fast retransmit with an neat flow diagram.	CO2	(06)
4.	a)	Describe the complete working of TCP connection for the given Fig.4(a).	CO2	(08)

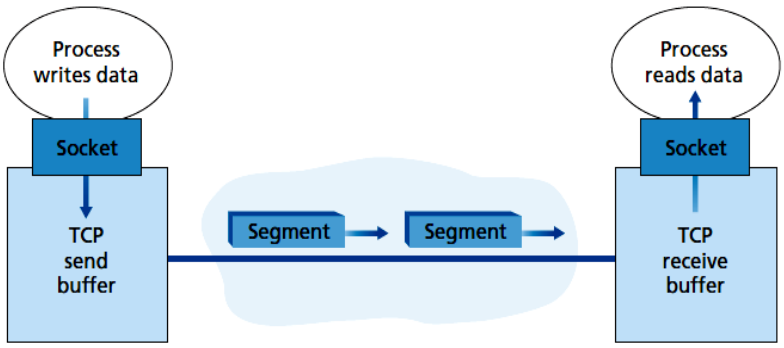

Fig. 4(a)

	b)	Explain congestion avoidance mechanism with respect to TCP congestion control.	CO2	(06)
	c)	Suppose that the three measured Sample RTT values are 106 ms, 120 ms, 140 ms. Compute the Estimated RTT after each of these Sample RTT values is obtained, using a value of α= 0.125 and assuming that the value of Estimated RTT was 100 ms just before the first of these three samples were obtained.	CO2	(06)

UNIT - III

5.	a)	Explain how routing tables are updated using Routing Information protocol.	CO3	(06)
	b)	An organization is granted a block of addresses with the beginning address 195.1.2.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.	CO3	(06)
	c)	Discuss IPv4 datagram format.	CO3	(08)
6.	a)	Explain Dijkstra's algorithm for shortest path calculation of routes in a network.	CO3	(06)
	b)	An ISP is granted the block 80.70.56.0/21. The ISP needs to allocate addresses for two organizations each with 500 addresses, two organizations each with 250 addresses, and three organizations each with 50 addresses. i. Find the number and range of addresses in the ISP block. ii. Find the range of addresses for each organization.	CO3	(06)
	c)	Discuss IP fragmentation with suitable example.	CO3	(08)

UNIT- IV

7.	a)	Differentiate between bit-oriented and character-oriented framing with suitable examples.	CO4	(06)
	b)	Given the dataword 101001111 and the divisor 10111, show the generation ofthe CRC codeword at the sender site (using binary division) and compute the decoding on receiving side to check errors.	CO4	(08)
	c)	Draw the flow diagram for CSMA/CD method.	CO4	(06)
8.	a)	With neat diagram explain the process of encoding and decoding in CRC.	CO4	(08)
	b)	Given the dataword1001 and the divisor 1011, show the process of encoding and decoding at the sender site and receiver site using polynomial division.	CO4	(08)
	c)	What are random access protocols. Explain two features of random-access protocols.	CO4	(04)

UNIT - V

9.	a)	Explain three causes of signal impairments with an example.	CO5	(08)
	b)	Write the Comparison ASK, FSK, PSK with suitable example.	CO5	(06)
	c)	With a neat block diagram explain Pulse Code Modulation (PCM) method.	CO5	(06)
10.	a)	Illustrate Nyquist Bit Rate and Shannon's Capacity with suitable formulas.	CO5	(08)
	b)	If the signal at the beginning of a cable with −0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?	CO5	(04)
	c)	Discuss the three categories of Analog-to-Analog signal conversion with an example.	CO5	(08)

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